

Mawlana Bhashani Science and Technology University

Department of Information and Communication Technology

3rd year 2nd Semester B. Sc. (Engg.) Final Examination 2024

Course Title: Web Application Development

Full Marks: 70

Course Code: ICT 3203

Time: 3 Hours

Answer any five from the following questions

1. (a) Define web engineering. 2
(b) Write down the characteristics of web application. 3
(c) Why web engineering is important to introduce umbrella method in developing web application? 4
(d) Describe why google.com follows all characteristics of web application? 5
2. (a) Describe the role of an access network in a communication system and discuss five (05) major types of access networks. 5
(b) What is HTML tags? Give some example of tags that are used in HTML table. 4
(c) Design a 4×4 HTML table to display structured data using appropriate table tags. The table must include the use of both *rowspan* and *colspan* attributes, and the table cells should be properly filled with relevant data. 5
3. (a) What is CSS? Why CSS is used in web application development? 4
(b) Compare inline, internal, and external CSS methods and analyze which method is most suitable for large-scale web applications. 6
(c) Given that $A \{ B: C \}$. According to CSS identity and define A, B, and C. 4
4. (a) Explain and give coding example of class selector, id selector. 4
(b) Create a full-screen CSS Grid where a 4-column layout features a header and footer spanning all columns, a sidebar spanning two rows on the left, and a "Main Profile" area situated above two smaller "Left Side" and "Right Side" sections. Use semantic HTML and grid-template-areas to map these specific regions and apply distinct background colors like red, yellow, green, and blue to each. 6
(c) What is Navigation bar in website? Why Navigation bar is used? 4
5. (a) Discuss some tips of CSS performance optimization. Write code if necessary. 4
(b) Discuss CSS media query with proper syntax. 4
(c) Design a responsive horizontal navigation bar using HTML and CSS, where the menu includes multiple top-level items such as Home, About, Project, News and Contact. Style the navigation bar using a yellow background with gray hover effects. Implement all styling using an external CSS file. 6
6. (a) "JavaScript can update and change both HTML and CSS, it can also calculate, manipulate, and validate data" – Justify the statement with coding. 6
(b) Create a simple web-based calculator using HTML, CSS, and JavaScript with two input fields for numbers and two buttons: a green Add button and a red Subtract button. When a button is clicked, the result should update immediately below the buttons. Style the calculator neatly, and treat any blank input as zero. 6
(c) List four local server stacks used for web development and briefly mention the platform each one supports. 2
7. (a) Create a program using asynchronous JavaScript that displays basic student info immediately and fetches their results with a delay, showing them once loaded. 5
(b) What is AJAX? How does AJAX work? 4
(c) Explain JSON, its purpose, syntax, and structure. Create a sample JSON file for a student containing basic info, courses, and friends as arrays of objects, and demonstrate how to access a specific value from it using JavaScript. 5

8. (a) "PHP (Hypertext Preprocessor) is a general-purpose scripting language that is mainly used for server-side scripting." Justify this statement by creating a form where the user inputs data, and the server stores (JSON file) or processes it using PHP. Include both the HTML form and the PHP processing script in your answer. 5
- (b) Explain API with its types in web application development. 4
- (c) You are tasked with developing a JavaScript-based feature for a web application that verifies the validity and deliverability of a user's email address before allowing them to proceed with registration. To accomplish this, you will use the Abstract Email Validation API, which provides detailed information about an email address, including its deliverability status. Write a JavaScript program that uses the Fetch API to send a GET request to the API endpoint <https://emailvalidation.abstractapi.com/v1/>, passing the API key `1x38e6ff3cae4458bf885301abc5a21x` and the email address `xyz@gmail.com` as query parameters. The program should then parse the JSON response, check the deliverability field, and print "The EMAIL is deliverable" if the field's value is 'The Email is Deliverable'. Otherwise, it should print 'The EMAIL is Not Deliverable.' 5